23. Friday, August 9. Compact subspaces of $\mathbb{R}^n$

Last Time: For any $a<b$, The interval $[a,b]$ is compact

Recall from Analysis:

Extreme Value Theorem: If $f:[a,b] \longrightarrow \mathbb{R}$ is a continuous map, Then There exist $x_{min}, x_{max} \in [a,b]$ such That $f(x_{min}) \leq f(x) \leq f(x_{max}) \quad \forall x \in [a,b]$.

Now we have a generalized version:

The Extreme Value Theorem: Assume $f: X \longrightarrow \mathbb{R}$ is a continuous map where X is a compact space. Then There exist $x_{min}, x_{max} \in X$ such That

$$f(x_{min}) \leq f(x) \leq f(x_{max}) \quad \forall x \in X$$

We will need The following:

Lemma 1: If $Y \subseteq \mathbb{R}$ is a compact space, there exists $M \in Y$ such that $y \leq M \quad \forall y \in Y$.

Proof: Assume for a contradiction that for every $M \in Y$ there exists some $y \in Y$ such that $M < y$. Then we have $Y \subseteq \bigcup_{y \in Y} (-\infty, y)$. Since Y is compact, we know that there exist $y_1, \ldots, y_m \in Y$ such that $Y \subseteq (-\infty, y_1) \cup \cdots \cup (-\infty, y_m)$

Thus, if we take $y^* = \max\{y_1, \ldots, y_m\}$, we get $Y \subseteq (-\infty, y^*)$

However, $y^* \in Y$, but $y^* \notin (-\infty, y^*)$, a contradiction $\square$

Proof of the Extreme Value Thm: Since $f: X \to \mathbb{R}$ is continuous and X is compact, we know that $f(X) \subseteq \mathbb{R}$ is compact

By lemma 1, we know that there exists $M \in f(X)$ such that $y \leq M \; \forall y \in f(X)$

Since $M \in f(X)$, there exists $x_{max} \in X$ such that $M = f(x_{max})$

Now, for every $x \in X$, since $f(x) \in f(X)$, we know that

$f(x) \leq M$, in other words, $f(x) \leq f(x_{max})$.

The existence of x_{min} follows similarly. $\square$

Definition: A subset $X \subseteq \mathbb{R}^n$ is called **bounded** if there exists $R \in \mathbb{R}$ such that $\|x\| \leq R \quad \forall x \in X$. In other words, if $X \subseteq B(0, R)$.

Theorem 2: A subspace $X \subseteq \mathbb{R}^n$ is compact if and only if X is closed and bounded.

Proof: ($\Rightarrow$) Assume $X \subseteq \mathbb{R}^n$ is compact. Since $\mathbb{R}^n$ is Hausdorff, we know from Prop 20.4 that X is closed. Also, since $X \subseteq \mathbb{R}^n = \bigcup_{R>0} B(0, R)$ and X is compact we know that $\exists\ R_1, \ldots, R_k$ such that $X \subseteq B(0, R_1) \cup \cdots \cup B(0, R_k)$

But if we let $R^* = \max\{R_1, \ldots, R_k\}$, we obtain $X \subseteq B(0, R^*)$, thus X is bounded, as desired.

($\Leftarrow$) Assume now $X \subseteq \mathbb{R}^n$ is closed and bounded.

Since X is bounded, we have, $X \subseteq B(0,R)$ for some $R>0$.

But $B(0,R) \subseteq [-R,R]^n$, so $X \subseteq [-R,R]^n$. Since $[-R,R]$ is compact we know that $[-R,R]^n$ is compact. Therefore $X \subseteq [-R,R]^n$ is closed in a compact space.

Therefore X is compact by Prop 20.3. $\square$

Compact Metric spaces

Definition: Let (X,d) be a metric space. Let $A \subseteq X$ be a non-empty subset.

For each $x \in X$, we define

$$d(x,A) = \inf \{ d(x,a) \mid a \in A \}$$

Note: 1) This inf exists because the set $\{ d(x,a) \mid a \in A \}$ is non-empty and 0 is a lower bound.

2) This defines a function $d(-,A) : X \longrightarrow \mathbb{R}$, $x \longmapsto d(x,A)$.

Proposition 3: The function $d(-, A): X \to \mathbb{R}$ is continuous.

Proof: Exercise!

Definition: We say that a subset $\emptyset \neq A$ of a metric space (X, d) is **bounded** if $\exists\ M > 0$ such that $d(a, b) \leq M \quad \forall\ a, b \in A$.

- If A is bounded, we define the **diameter of A** as

$$\text{diam}(A) = \sup\{d(a,b) \mid a, b \in A\}$$

Note: This sup exists because $\{d(a,b) \mid a, b \in A\}$ is non-empty and has an upper bound.

The Lebesgue number Lemma: Let (X, d) be a compact metric space and let $\mathcal{U}$ be an open covering of X. Then there exists $\delta > 0$ such that for every subset $A \subseteq X$ with $\text{diam}(A) < \delta$, there exists $U \in \mathcal{U}$ with $A \subseteq U$